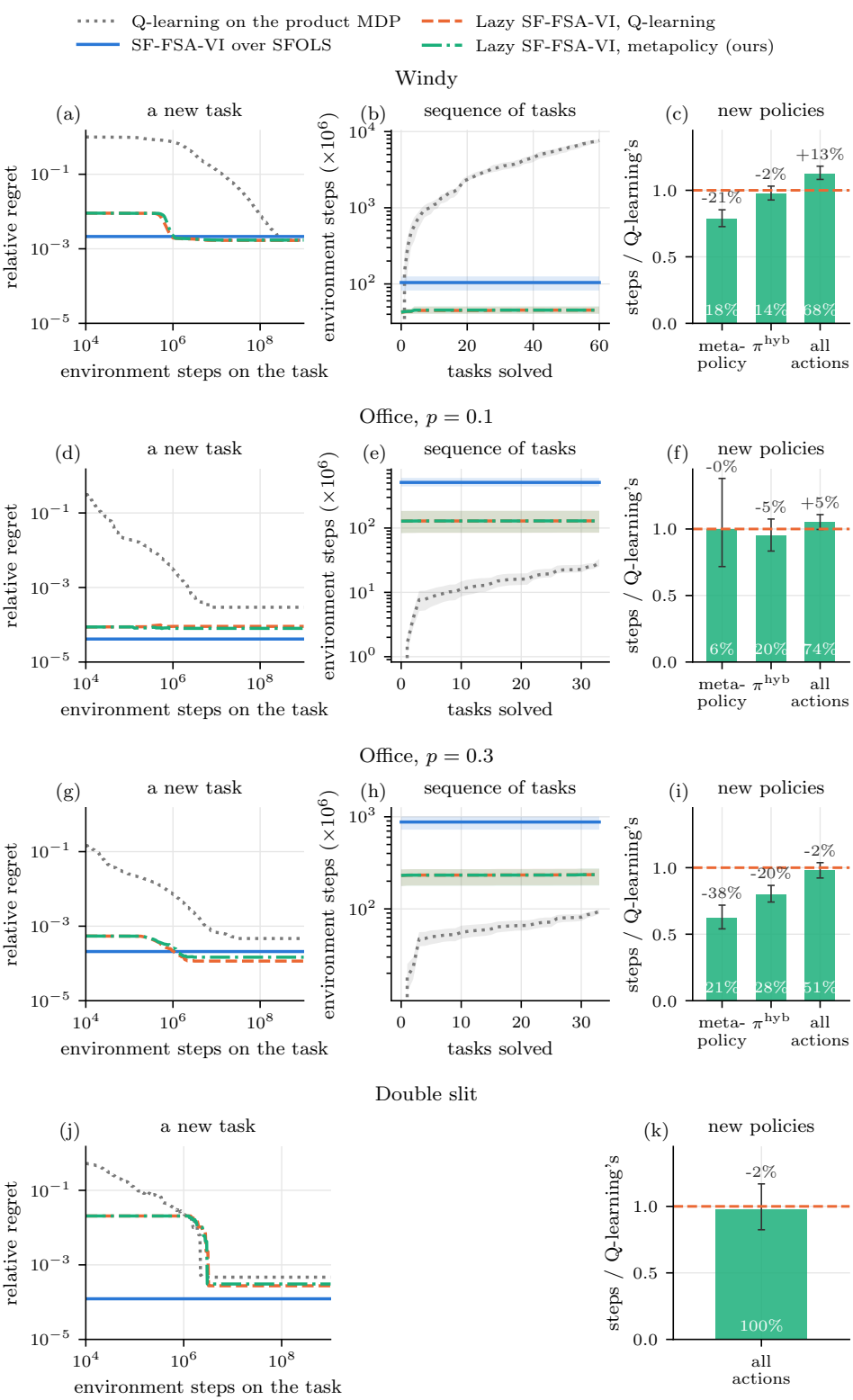

Q-learning on the product MDP
SF-FSA-VI over SFOLS
Lazy SF-FSA-VI, Q-learning
Lazy SF-FSA-VI, metapolicy (ours)
Windy
(a)
a new task
relative regret
environment steps on the task
(b)
sequence of tasks
environment steps (×10^6)
tasks solved
(c)
new policies
steps / Q-learning's
-21%
-2%
+13%
18%
14%
68%
meta-policy
π^hyb
all actions
Office, p = 0.1
(d)
a new task
relative regret
environment steps on the task
(e)
sequence of tasks
environment steps (×10^6)
tasks solved
(f)
new policies
steps / Q-learning's
-0%
-5%
+5%
6%
20%
74%
meta-policy
π^hyb
all actions
Office, p = 0.3
(g)
a new task
relative regret
environment steps on the task
(h)
sequence of tasks
environment steps (×10^6)
tasks solved
(i)
new policies
steps / Q-learning's
-38%
-20%
-2%
21%
28%
51%
meta-policy
π^hyb
all actions
Double slit
(j)
a new task
relative regret
environment steps on the task
(k)
new policies
steps / Q-learning's
-2%
100%
all actions